Forensics for Cybercrimes Committed Using Artificial Intelligence

PhD Research Presentation

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Presentation Outline

Introduction

Artificial Intelligence is increasingly being incorporated into systems used for communication, automation, decision-making and cyber operations.

This creates a new forensic challenge:

Research Problem

How can digital investigators reliably identify, collect, preserve, analyse and present evidence when cybercrimes involve AI-generated or AI-assisted activity?

Problem Statement

Traditional digital forensic approaches generally assume that evidence can be associated with identifiable devices, accounts, users or processes.

AI-assisted cybercrime introduces additional questions:

- Who actually performed the action?
- Was the action human-generated or AI-assisted?
- How can investigators establish provenance?
- Can AI-generated evidence be authenticated?
- How should attribution be established?
- How can such evidence be presented in court?

Research Objectives

General Objective

Objective

To develop a digital forensic framework for investigating cybercrimes committed using Artificial Intelligence.

Specific Objectives

1. Examine existing AI-assisted cybercrime techniques.
2. Evaluate existing digital forensic approaches.
3. Investigate methods for establishing evidence provenance and integrity.
4. Develop a forensic framework for AI-assisted cybercrime investigations.
5. Evaluate the proposed framework.

Research Questions

1. How is AI being used to facilitate cybercrime?
2. What limitations exist in current digital forensic approaches?
3. How can AI-generated evidence be authenticated?
4. How can provenance and attribution be established?
5. What forensic framework can support investigation and courtroom presentation?

Literature Review

The literature will be examined across four major areas:

Digital Forensics

- Evidence acquisition
- Preservation
- Analysis
- Presentation

AI and Cybercrime

- Generative AI
- Deepfakes
- Automated attacks
- AI-assisted fraud

Theoretical Framework

The study will examine relevant theories and formal models that can explain forensic evidence and attribution.

Potential theoretical foundations include:

- Digital Forensics Theory
- Information Security Theory
- Evidence and Authentication Principles
- Attribution Theory
- Computational and Formal Reasoning

Research Methodology

Research Approach

The study will employ an appropriate mixed-methods / computational research approach depending on the research phase and research questions.

- Literature analysis
- Digital forensic experimentation
- Dataset analysis
- AI-assisted cybercrime scenarios
- Framework development
- Experimental evaluation

Proposed Forensic Framework

AI Cybercrime Forensic Process

Collection → Preservation → Authentication → Analysis → Attribution → Presentation

Expected Contribution to Knowledge

The research aims to contribute:

1. A forensic framework for AI-assisted cybercrime.
2. Methods for assessing provenance of AI-generated digital evidence.
3. Approaches for authentication and integrity assessment.
4. A structured approach to AI-assisted cybercrime attribution.
5. Guidance for forensic investigators and legal practitioners.

Conclusion

Central Argument

AI changes not only how cybercrimes can be committed, but also how digital evidence must be interpreted, authenticated and attributed.

The research therefore seeks to bridge the gap between:

$$\boxed{\text{Artificial Intelligence} \quad + \quad \text{Cybercrime} \quad + \quad \text{Digital Forensics} \quad + \quad \text{Evidence}}$$

References

- Author, A. (2025). *Digital Forensics and Artificial Intelligence*. Journal of Cybersecurity Research.
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Thank You

Questions and Discussion

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